SOHelioraX

SolarShield AI

AI-Powered Solar Flare Nowcasting & Risk Forecasting System

Built on Real ISRO Aditya-L1 Data (SoLEXS + HEL1OS)

PROJECT REPORT

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Abstract

Solar flares release intense bursts of X-ray and energetic-particle radiation that can disrupt satellites, GPS, aviation, power grids and communication networks. SOHelioraX is an end-to-end space-weather intelligence platform that detects, classifies and forecasts solar flares in real time using data from India's first dedicated solar mission, Aditya-L1. It fuses the SoLEXS (Soft X-ray) and HEL1OS (Hard X-ray) Level-1 instruments with a trained Random Forest machine-learning model to produce flare classifications (A/B/C/M/X), a four-level operational risk score, short-term probability forecasts, and early-warning alerts — all delivered through an interactive web dashboard. The model, trained on 739 days of real observations, achieves a cross-validated accuracy of 97.7% ± 1.1%.

1. Introduction

Space weather, driven primarily by solar activity, has a direct and growing impact on modern technological infrastructure. Solar flares are among the most significant space-weather events, capable of causing radio blackouts, satellite anomalies and navigation errors within minutes of eruption. India's Aditya-L1 mission, positioned at the Sun-Earth L1 point, continuously observes the Sun and provides valuable X-ray data through its on-board payloads.

However, most operational space-weather systems rely on foreign data sources (NOAA/GOES) and fixed thresholds, and Aditya-L1 Level-1 data is largely used for post-event scientific analysis rather than real-time operational warning. SOHelioraX addresses this gap by transforming raw Aditya-L1 telemetry into instant, human-readable, actionable intelligence.

2. Objectives

- Build an India-built, real-time solar-flare nowcasting and forecasting system using Aditya-L1 data.
- Automatically read and process SoLEXS and HEL1OS Level-1 FITS files.
- Classify solar flares (A/B/C/M/X) using a machine-learning model combined with physics-based logic.
- Compute a four-level operational risk score (Green/Yellow/Orange/Red) with confidence values.
- Provide short-term (1H/2H/4H) probability forecasts and an estimated early-warning lead time.
- Present everything through an interactive dashboard with alerting capability.

3. How It Differs From Existing Systems

- Existing tools rely on NOAA/GOES data and fixed thresholds; SOHelioraX uses India's own Aditya-L1 mission.
- Aditya-L1 data is mostly used post-event — our system fuses live SoLEXS + HEL1OS streams for real-time use.
- Combines physics-based classification with a trained ML model (hybrid approach) for robustness.
- First integrated Aditya-L1 prediction dashboard with nowcasting, forecasting, risk gauges and alerts.

4. Dataset

The dataset is built from real Aditya-L1 Level-1 data downloaded from the ISSDC PRADAN portal, covering 739 days of observations. Two instruments are used: SoLEXS (Soft X-ray Spectrometer) and HEL1OS (High Energy L1 Orbiting X-ray Spectrometer).

Total observation days: 739

Instruments: SoLEXS (Soft X-ray) + HEL1OS (Hard X-ray)

Features per day (8): SoLEXS peak, mean, std, median, p95 + HEL1OS peak, mean, std
Labeling: Flare class from SoLEXS soft X-ray peak (GOES standard); zero/missing days skipped

Flare classification thresholds (SoLEXS soft X-ray peak counts):

Class	Threshold	Meaning
A	< 150	Quiet / very low
B	< 750	Minor
C	< 3,500	Moderate
M	< 18,000	Strong
X	≥ 18,000	Extreme (rare)

5. Methodology & System Architecture

The system follows a four-layer architecture:

- Data Layer — raw Aditya-L1 FITS files (SoLEXS & HEL1OS Level-1) from ISSDC PRADAN.
- Processing Layer — data ingestion, preprocessing (noise reduction, missing-value handling, normalization) and feature engineering (peak, mean, std, median, p95).
- AI / Logic Layer — peak detection, Random Forest classifier and risk forecaster.
- Presentation Layer — Flask web application, interactive dashboard and alert system.

End-to-end flow: Raw Space Data → Process & Prepare → AI Analysis & Prediction → Dashboard & Visualization → Alerts & Insights.

6. Machine Learning Model

A Random Forest Classifier (300 trees, balanced class weights) is trained on the engineered features to classify flares into A/B/C/M/X. The model is validated using honest 5-fold Stratified Cross-Validation, reporting mean accuracy with standard deviation rather than a single optimistic number. Class probabilities from the model are blended 50/50 with physics-based (peak-derived) probabilities to improve robustness, and a secondary Random Forest Regressor estimates a continuous peak-based risk score.

Risk Level	Flare Class	Peak Range
Green	A / B	< 750
Yellow	C	750 – 3,500
Orange	M	3,500 – 18,000
Red	X	18,000+

7. System Implementation

The backend is built with Flask (Python). Trained models are persisted with joblib and loaded at start-up. Astropy reads the FITS files, and the prediction logic outputs the flare class, C/M/X probabilities, confidence, risk level and 1H/2H/4H forecasts. The frontend is a dark, dashboard-style interface featuring a live nowcasting panel, dual risk gauges, a live SoLEXS light curve, a 739-day activity trend, a flare-class distribution, a month-wise 739-day data explorer, and an email/SMS alert subscription. The application is self-healing: if live SoLEXS files are unavailable, it automatically falls back to committed sample data so the dashboard never crashes.

8. Results

- Cross-validated model accuracy: 97.7% ± 1.1% (5-fold stratified).
- Trained on 739 days of real Aditya-L1 observational data.
- Realistic flare-class distribution (X-class rare, B/C common) after log-threshold calibration.
- Four-level operational risk scoring with confidence values.
- 1H / 2H / 4H probability forecasts with an estimated ~30-minute early-warning lead time.
- Interactive dashboard with live nowcasting, historical explorer and alerting.

9. Limitations (Transparent Engineering)

- The deployed live panel shows committed sample SoLEXS data, not a real-time feed, as a public live Aditya-L1 stream is not available; all other components run on real data.
- The ~30-minute lead time is a heuristic estimate based on flare rise time, not a validated forecasting metric.
- Email alerts are fully functional (free Gmail app-password); real SMS requires a paid gateway (e.g., Twilio/Fast2SMS).
- Flare class is peak-defined per the GOES standard; the model classifies from the full statistical profile of SoLEXS + HEL1OS.

10. Deployment

The application is deployed on Render as a web service using Gunicorn (gunicorn app:app). The committed dataset (CSV/JSON), trained models (.pkl) and sample data allow the app to run on the cloud without needing the raw Aditya-L1 archive, making the deployment lightweight and reproducible.

11. Conclusion & Future Scope

SOHelioraX demonstrates a practical, India-built pipeline that turns Aditya-L1 Level-1 data into real-time, actionable space-weather intelligence with strong, honestly-validated accuracy. Future work includes integrating a genuine real-time data feed, validating lead-time predictions against event catalogs, adding energetic-particle and CME indicators, enabling production SMS alerting, and extending the model with deep-learning time-series methods for longer-horizon forecasting.

References

- ISRO Aditya-L1 Mission — SoLEXS & HEL1OS payloads.
- ISSDC PRADAN Portal — Aditya-L1 Level-1 data archive.
- GOES Solar X-ray Flux — flare classification standard (NOAA).
- scikit-learn, Astropy, Flask, pandas, NumPy, Matplotlib — open-source libraries.